PROJECT REPORT

Module 3 - AI-Based Automotive Review and Customer Sentiment Analysis

Submitted by

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1. Project Overview

This project implements a complete software prototype for automotive review analytics. A user can enter one review or upload a CSV of multiple reviews. The system cleans the text, detects automotive aspects, runs Transformer-based sentiment classification and presents the results through a Streamlit dashboard.

2. Objectives

Objective	Implementation
Analyze automotive reviews	Single-review and batch CSV workflows
Identify aspect-level sentiment	Controlled aspect vocabulary + local clause context
Use BERT / XLM-R	Configurable Hugging Face Transformer models
Create usable analytics	Streamlit dashboard + downloadable CSV
Support offline demonstration	Deterministic lexicon fallback when model weights are unavailable

3. System Architecture

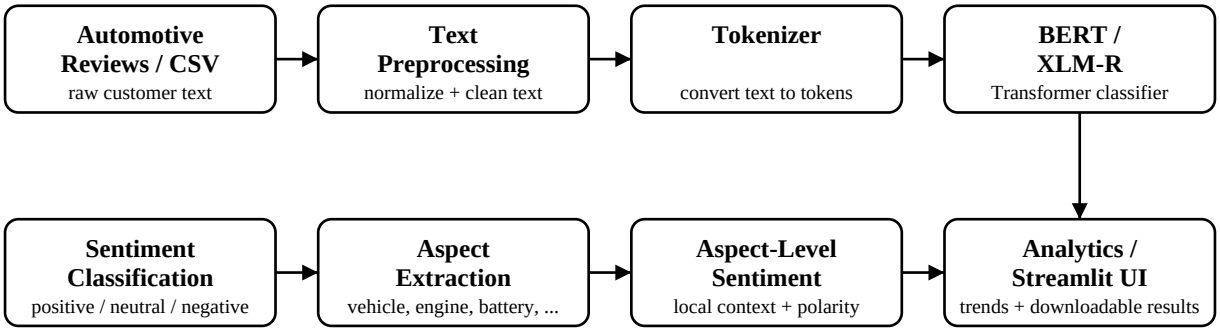

Figure 1. System architecture - review ingestion through aspect-level sentiment analytics.

4. Software Modules

Module	Responsibility
src/preprocess.py	Text normalization and clause splitting
src/aspects.py	Automotive aspect vocabulary and local context selection
src/sentiment_engine.py	XLM-R primary model, BERT alternate model and fallback classifier
src/analyzer.py	End-to-end review analysis and result flattening
src/analytics.py	Batch DataFrame processing and aspect summaries
src/cli.py	Command-line batch analysis
app/app.py	Interactive Streamlit interface
tests/test_core.py	Smoke tests for preprocessing, aspects and offline analysis

5. Aspect Extraction Method

The project uses a controlled vocabulary based directly on the nine automotive aspects in the brief. Each aspect contains representative keywords. Reviews are split into sentence/contrast clauses, and each detected aspect is attached to the local clause where its keyword occurs. This keeps the design explainable and makes it possible to show the context sent to the sentiment classifier.

6. Sentiment Classification

The primary configuration loads CardiffNLP's XLM-R sentiment model through the Transformers pipeline. The project also supports the multilingual BERT product-review sentiment model. The engine maps model output labels to a common Positive / Neutral / Negative representation, allowing the dashboard to work with either model. The Hugging Face documentation confirms that text-classification pipelines are built around fine-tuned sequence-classification models. [1]

7. Fallback and Reliability Design

Model weights can be large and the first model-backed run may require internet access. To keep the application demonstrable in classrooms or restricted environments, the code catches model-loading and inference failures and falls back to a small lexicon-based scorer. The fallback is explicitly labeled in the output as offline-fallback; it is not presented as equivalent to the Transformer model.

8. Example Workflow

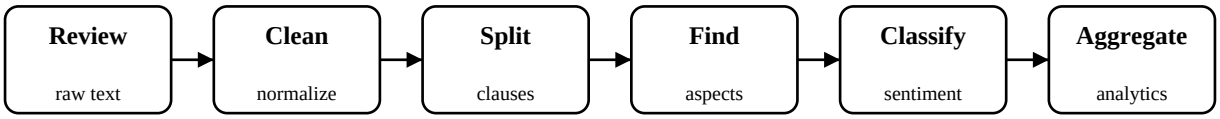

Figure 2. End-to-end workflow.

Example input: “The battery range is excellent, but the charging time is too long.” The analyzer separates the contrasting clauses, associates each clause with the relevant aspect and returns opposite polarity labels for the two aspects.

9. Demonstration Dataset

The source code includes a 24-review CSV containing examples across the nine aspects. It is deliberately labeled as demonstration data rather than a benchmark dataset. The purpose is software validation: file loading, aspect extraction, batch processing, result serialization and dashboard display.

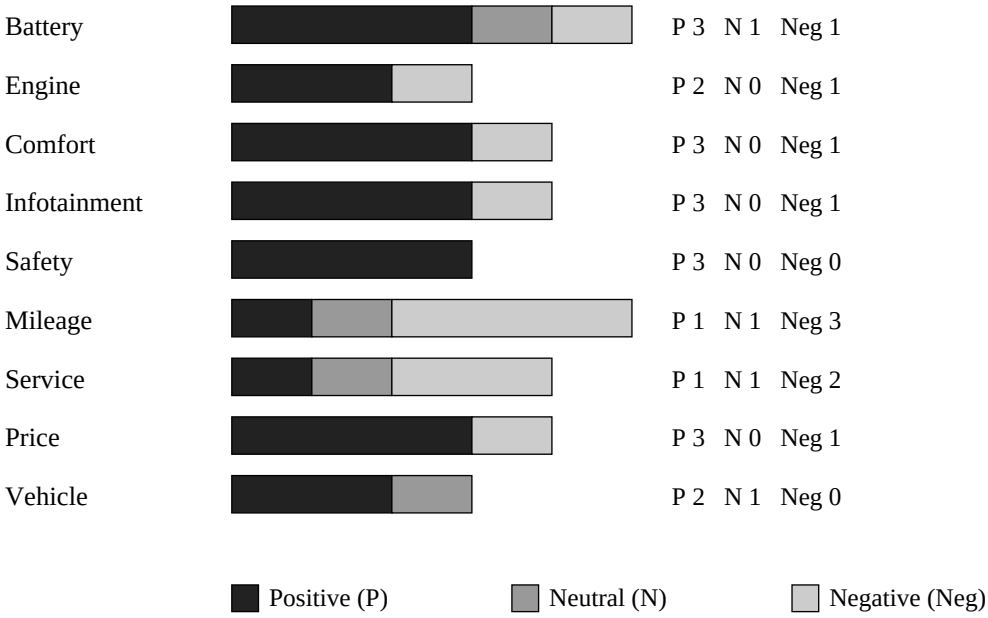

Figure 3. Illustrative aspect polarity counts on the bundled demo set; these counts are not model accuracy measurements.

10. Results Produced by the Application

Output field	Example
Overall sentiment	Positive / Neutral / Negative
Overall confidence	Model score returned by classifier
Aspect	Battery, Engine, Safety, etc.
Aspect sentiment	Positive / Neutral / Negative
Aspect confidence	Classifier score for local context
Context	Local clause supporting the aspect decision
Backend	transformers or offline-fallback
Model	XLM-R or BERT model identifier

11. Installation and Execution

Step 1. Create and activate a Python virtual environment. Step 2. Install dependencies with `pip install -r requirements.txt`. Step 3. Launch the dashboard with `streamlit run app/app.py`. Step 4. Select XLM-R, BERT or Offline demo mode. Step 5. Analyze a review or upload a CSV.

The command-line path is `python -m src.cli --input data/sample_reviews.csv --output outputs/automotive_sentiment_results.csv`.

12. Testing and Evaluation Protocol

The source package contains smoke tests covering preprocessing, aspect detection and offline analysis. For a formal evaluation, use an independently labeled automotive review dataset and report class-wise precision, recall, F1 and confusion matrix. For the aspect stage, annotate aspect presence and polarity separately. Also report latency and throughput on the actual deployment machine.

13. Limitations

The current aspect extraction is vocabulary-driven and therefore may miss synonyms, spelling variants or unseen automotive terminology. Local-clause attribution is a transparent approximation to full aspect-based sentiment analysis. The pretrained sentiment models are general-domain models, so automotive-domain performance should be validated before operational use.

14. Future Scope

Future work can replace the vocabulary stage with a trained aspect extractor, fine-tune a Transformer on labeled automotive reviews, add multilingual Indian-language review support, introduce topic clustering, expose trends over time and compare BERT vs XLM-R empirically on the same held-out dataset.

15. Conclusion

The project turns the Module 3 concept into a runnable end-to-end software system. It combines NLP preprocessing, aspect extraction, Transformer sentiment classification and an analytics interface. The design is modular, testable and explicit about the difference between model-backed inference and the offline demonstration fallback.

References

[1] Hugging Face Transformers documentation - sequence classification and pipelines.

[2] CardiffNLP, twitter-XLM-RoBERTa-base for Sentiment Analysis, Hugging Face model card.

[3] NLP Town, bert-base-multilingual-uncased-sentiment, Hugging Face model card.